```
abstract class EnergySource {
    String sourceId;
    String sourceName;
    double energyGenerated;

    EnergySource(String sourceId, String sourceName, double energyGenerated) {
        this.sourceId = sourceId;
        this.sourceName = sourceName;
        this.energyGenerated = energyGenerated;
    }


    abstract double calculateEfficiency();


    void displayDetails() {
        System.out.println("Source ID: " + sourceId);
        System.out.println("Source Name: " + sourceName);
        System.out.println("Energy Generated: " + energyGenerated + " kWh");
        System.out.printf("Efficiency: %.2f\n", calculateEfficiency());
        System.out.println("----------------------------------");
    }
}


class SolarEnergy extends EnergySource {

    SolarEnergy(String sourceId, String sourceName, double energyGenerated) {
        super(sourceId, sourceName, energyGenerated);
    }


    double calculateEfficiency() {
        return (energyGenerated / 5000.0) * 100.0;
    }
}


class WindEnergy extends EnergySource {

    WindEnergy(String sourceId, String sourceName, double energyGenerated) {
        super(sourceId, sourceName, energyGenerated);
    }
```

```
    double calculateEfficiency() {
        return (energyGenerated / 8000.0) * 100.0;
    }
}
public class EnergyManagementSystem {
    public static void main(String[] args) {

        EnergySource ref;

        ref = new SolarEnergy("101", "Solar Park", 2560.0);
        ref.displayDetails();


        ref = new WindEnergy("202", " Wind Energy", 4700.0);
        ref.displayDetails();
    }
}
```

```
 ----jGRASP exec: java EnergyManagementSystem
Source ID: 101
Source Name: Solar Energy
Energy Generated: 2560.0 kWh
Efficiency: 51.20
-----------------------------------
Source ID: 202
Source Name: Wind Energy
Energy Generated: 4700.0 kWh
Efficiency: 58.75
-----------------------------------

 ----jGRASP: Operation complete.
```

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